Aside from the mind-blowing-ness of it, the CMB gives us an invaluable window into the universe's first moments, and into how it has grown and evolved over time. It also gives us some hints about where it is all going, as we'll see in later chapters.

That said, when you make a map of the CMB that shows the color variation of the light across the sky, it actually looks fairly boring: it is *very nearly* exactly the same color everywhere. The minute deviations you can detect, though, as tiny as they are, tell us a lot. When they crank up the contrast enough to get some color variation, astronomers can see that the CMB looks ever so slightly blotchy, as if someone did an abstract pointillism painting on the sky with a brush as big around as the full Moon viewed from Earth. These blotches collect in clumps of one color in some places and mix in others, with some of the spots a little tiny bit redder, some a little tiny bit bluer.[fn12] The color variations reveal places where the roiling primordial cosmic plasma was just slightly cooler or hotter, due to very, very slight changes in density—each point has a density that deviates from the average by no more than about one part in 100,000. (For some perspective on what one part in 100,000 looks like, empty a soda can into a backyard swimming pool.)

We can, with careful calculations, work out how those tiny variations in density are destined to grow over time, starting from minuscule blips and, over the millennia, growing up into entire clusters of galaxies. Gravitational collapse is a powerful thing. If you have a little bit of matter that's denser than the matter around it, it will pull in more from those less dense places, which increases the contrast, which pulls in more, and so on. The rich get richer and the poor get poorer.

Using computer simulations that show billions of years passing in the span of seconds, we can watch as a patch of matter that is only a tiny fraction denser than the one next to it pulls in enough of its surrounding gas to form the first star in the universe. These stars form within the first galaxies, which gather into clusters of galaxies, which en masse turn the splotchy patchwork of the CMB into what we now see as a cosmic web: a veiny arrangement of nodes and filaments and voids traced out by the galaxies shining along it, like dew drops on a spider's web. If you compare the results of one of these simulations with an actual map of the cosmos, where each galaxy is one point in a giant 3D map, they are in such incredible agreement that you won't be able to tell the difference.